Adaptive KDE Bandwidth Selection for Overlap Loss in Semi-Supervised Anomaly Detection*

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Abstract

Anomaly Detection with Score Distribution Discrimination (ADSD) uses Overlap Loss to separate the score distributions of unlabelled observations and labelled anomalies. This study investigates a mini-batch KDE bandwidth estimate based on Silverman's rule and its interpolation with the original fixed bandwidth. Experiments cover Thyroid, Seismic Bumps, and a constructed synthetic dataset, using three labelled-anomaly ratios, five bandwidth weights, and five random seeds, giving 225 runs. At 5% supervision, the fixed-bandwidth Thyroid replication achieved a mean ROC-AUC of 0.9811 and average precision (AP) of 0.8138, closely matching the published results. Bandwidth effects varied across datasets and supervision levels. On the synthetic dataset, $\alpha = 0.8$ achieved the highest mean AP at all three ratios, but paired comparisons with the fixed baseline were not statistically significant at the 5% level. The results do not establish a consistent advantage for the tested adaptive bandwidth method.

1 Introduction

Anomaly detection is used to identify unusual observations that differ from expected patterns in a dataset. It has applications in financial fraud detection, network security, and industrial monitoring. However, detecting anomalies remains challenging because abnormal observations are often rare, and obtaining reliable labels can be difficult (Chandola et al., 2009). In semi-supervised anomaly detection, a small number of labelled anomalies may be available alongside a much larger set of unlabelled observations.

Anomaly Detection with Score Distribution Discrimination (ADSD) addresses this problem by learning to separate the anomaly score distributions of labelled and unlabelled data (Jiang et al., 2023). The method uses Overlap Loss, which measures the overlap between score distributions estimated through kernel density estimation (KDE). By reducing this overlap during training, the model aims to assign different scores to labelled anomalies and unlabelled observations.

Bandwidth selection is an important part of KDE because it controls the smoothness of the estimated probability density. A bandwidth that is too large can hide important differences between distributions, while a very small bandwidth can produce unstable estimates (Silverman, 1986). The original ADSD implementation uses a fixed KDE bandwidth for its MLP configuration. However, the spread of anomaly scores can vary between datasets and during model training, which raises the question of whether a fixed bandwidth is always suitable.

This study examines the effect of adaptive KDE bandwidth selection on ADSD performance. The original fixed-bandwidth method is first reproduced using the authors' implementation. An extension based on Silverman's bandwidth selection rule is then evaluated. A hybrid approach is used to combine the fixed bandwidth with an adaptive estimate at different interpolation levels.

Experiments are conducted on three datasets: Thyroid, Seismic Bumps, and a constructed synthetic dataset containing global, local, dependency, and cluster anomalies. Each configuration is evaluated across five random seeds using ROC-AUC and average precision (AP). These metrics provide complementary information, particularly when datasets contain relatively few anomalies (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015).

The main contribution of this study is a controlled comparison of fixed and adaptive KDE bandwidth selection within the ADSD framework. The results show whether adaptive bandwidth selection improves detection performance across differ-

*Source code, datasets, experimental results, and reproducibility materials are available at https://github.com/mustafaaats/ADSD-Adaptive-KDE.

ent datasets and highlight its limitations under the tested conditions.

2 Background and Related Work

2.1 Source Paper and Project Justification

The source paper for this project is *Anomaly Detection with Score Distribution Discrimination* by Jiang et al. (2023), published at the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining.[1] The paper introduces Overlap Loss, which trains an anomaly scoring model by separating the score distributions of labelled anomalies and unlabelled observations. Its evaluation covers 25 real-world datasets, four neural network architectures, and labelled-anomaly ratios of 5%, 10%, and 20%. Performance is measured using ROC-AUC and a precision–recall metric.

The paper was selected for several reasons. Publication at a peer-reviewed data mining conference provides an initial indicator of research quality. More importantly for this project, the authors provide source code and benchmark data, allowing the reported method to be investigated directly. The implementation also supports an MLP configuration that is practical to run within the project scope. A preliminary execution on Thyroid confirmed that the data and training pipeline could be used successfully.

The method also offers a clear opportunity for a controlled extension. The selected MLP configuration uses a fixed KDE bandwidth of 1.0. Replacing this value with a bandwidth estimated from the current scores allows one modelling choice to be examined while retaining the original architecture and training procedure. The project therefore investigates both the reproducibility of the selected baseline and the effect of bandwidth selection on new data.

2.2 Anomaly-Informed Detection

Traditional unsupervised anomaly detection methods identify unusual observations without using labelled anomalies. However, some applications provide a small number of confirmed abnormal cases. These labels can help a model recognise patterns that may be difficult to identify from unlabelled data alone. This setting is commonly described as anomaly-informed anomaly detection (Jiang et al., 2023).

[1] https://doi.org/10.1145/3580305.3599258

A common approach is to train a model to assign higher anomaly scores to abnormal observations. Some methods use predefined target scores or margins to separate normal and anomalous data. However, suitable targets can vary between datasets, particularly when only a few labelled anomalies are available. Jiang et al. (2023) identified this limitation as one motivation for developing Overlap Loss.

2.3 Overlap Loss and ADSD

Jiang et al. (2023) introduced Anomaly Detection with Score Distribution Discrimination (ADSD), which focuses on separating anomaly score distributions rather than forcing individual observations towards fixed target scores. The method estimates two distributions: one for unlabelled observations and another for labelled anomalies. Kernel density estimation (KDE) is then used to estimate their probability densities.

Overlap Loss measures the shared area between these estimated distributions. During training, the model attempts to reduce this overlap so that labelled anomalies receive scores that are easier to distinguish from the unlabelled data. An important feature of this approach is that the loss function can be applied to different neural network architectures, including multilayer perceptrons (MLPs), autoencoders, ResNet, and FTTransformer (Jiang et al., 2023).

The original study evaluated Overlap Loss on 25 datasets using different proportions of labelled anomalies. Its results showed that score distribution discrimination can be effective when anomaly labels are limited. The present study focuses on the MLP implementation to examine one specific part of the method: KDE bandwidth selection.

2.4 KDE Bandwidth Selection

KDE bandwidth controls the smoothness of an estimated density. Small values may produce unstable estimates, while large values may hide distributional differences (Silverman, 1986). Silverman's robust rule uses sample size, standard deviation, and interquartile range.

The selected ADSD MLP configuration uses a fixed bandwidth of 1.0. Since score distributions change across datasets and during training, this project compares that baseline with a mini-batch estimate and hybrid bandwidths while retaining the original architecture and training procedure.

3 Methodology

This study follows the original ADSD implementation and examines the effect of changing the KDE bandwidth used in Overlap Loss. The experimental approach consists of two stages. First, the original fixed-bandwidth model is reproduced. Second, an adaptive bandwidth method is introduced and compared with the baseline under the same training conditions.

3.1 Replication of ADSD

The replication uses the multilayer perceptron (MLP) architecture provided in the original ADSD implementation (Jiang et al., 2023). The model is trained for 20 epochs using stochastic gradient descent (SGD). The learning rate is 0.001, momentum is 0.7, and weight decay is 0.01. The batch size is 256.

The original resampling procedure is retained. Each training batch contains 128 unlabelled observations sampled without replacement and 128 labelled anomalies sampled with replacement. This allows the model to use a small number of known anomalies repeatedly during training. The unlabelled group also contains training anomalies whose labels have been hidden.

For the baseline, the KDE bandwidth is fixed at 1.0 for both the unlabelled and labelled anomaly score distributions. No changes are made to the MLP architecture or the original Overlap Loss objective.

Scope of replication. The replication focuses on Thyroid, the MLP architecture, and the 5% labelled-anomaly setting. This provides a direct comparison with a published result while keeping the experimental scope manageable. Thyroid was also used in the initial feasibility check, providing an established starting point for the controlled bandwidth comparison. The published ROC-AUC and precision–recall results for this configuration are reported in Tables A6 and A9 of the original supplementary material.

This experiment reproduces one configuration, rather than the complete evaluation of the source paper. It does not establish whether the original findings hold across all 25 datasets, other network architectures, or higher labelled-anomaly ratios. The extension experiments examine generalisation to Seismic Bumps and the constructed synthetic dataset under the selected configuration.

The MLP contains two hidden layers with 100 and 20 units, respectively, using ReLU activations. A linear layer produces a scalar score, followed by batch normalisation. The original network structure is retained for all bandwidth configurations.

3.2 Adaptive KDE Bandwidth Selection

The proposed extension replaces the fixed KDE bandwidth with a value calculated from the current anomaly scores. The bandwidth is estimated separately for the unlabelled and labelled anomaly distributions during each training mini-batch.

The adaptive method is based on Silverman's robust rule-of-thumb (Silverman, 1986). For a set of anomaly scores S containing n observations, the bandwidth is calculated as

$$h_{\text{raw}} = 0.9 \min\left(\sigma_S, \frac{\text{IQR}(S)}{1.34}\right) n^{-1/5}, \quad (1)$$

where σ_S is the sample standard deviation and $\text{IQR}(S)$ is the interquartile range.

To prevent extremely small or large bandwidth values, the estimate is restricted to the interval $[0.1, 1.0]$:

$$h_{\text{adapt}} = \text{clip}(h_{\text{raw}}, 0.1, 1.0). \quad (2)$$

The implementation also includes a fallback rule. If fewer than two scores are available, or if the estimated score dispersion is zero or non-finite, the bandwidth is set to 1.0. Bandwidth statistics are calculated without gradient tracking, so the bandwidth itself is not differentiated during model optimisation.

A single bandwidth is shared by all kernels within each score group. Adaptation therefore occurs between mini-batches and score groups, rather than assigning a different bandwidth to every observation. The clipping interval and fallback value are implementation choices of this project.

3.3 Hybrid Bandwidth Selection

A hybrid method is used to control the influence of adaptive bandwidth selection. The final bandwidth is calculated as

$$h_\alpha = (1-\alpha)h_{\text{fixed}} + \alpha h_{\text{adapt}}, \quad (3)$$

where $h_{\text{fixed}} = 1.0$ and $\alpha \in \{0, 0.2, 0.5, 0.8, 1.0\}$.

When $\alpha = 0$, the original fixed-bandwidth method is used. When $\alpha = 1$, the bandwidth is

fully adaptive. Intermediate values combine the two approaches.

The same interpolation weight is applied to both score distributions, although their adaptive bandwidth estimates are calculated separately. All other model and training settings remain unchanged.

Since the adaptive estimate is restricted to $[0.1, 1.0]$, the hybrid bandwidth can retain or reduce the baseline value of 1.0, but cannot increase it. The experiments therefore evaluate this restricted form of bandwidth adaptation.

3.4 Implementation and Reproducibility

Experiments were conducted using Python 3.10.20, PyTorch 1.13.1, NumPy 1.26.4, and scikit-learn 1.7.2. The selected MLP implementation uses the CPU. Dependency versions are provided in `requirements.txt`. Installation was tested in a fresh Python 3.10.20 virtual environment on macOS ARM64. Data integrity, synthetic-data reconstruction, and LLM scoring checks passed. Three representative 20-epoch experiments reproduced the archived ROC-AUC and AP values within 10^{-6}.

However, `pip check` reported a platform compatibility warning for PyTorch 1.13.1. The installed wheel metadata declares x86_64, whereas the inspected Torch binaries are ARM64. Imports and the tested training runs succeeded despite this metadata mismatch.

The bandwidth extension is implemented in `baseline/ADSD/fit.py`. The original model architecture and optimiser are retained. The repository contains the prepared datasets, the synthetic-data generator, experiment scripts, and individual run results.

The script `run_multiseed.py` evaluates the fixed bandwidth and the $\alpha = 0.2$ hybrid configuration at 5% supervision. The script `run_alpha_ablation.py` evaluates $\alpha = 0.5$, 0.8, and 1.0 at the same supervision level. The script `run_label_ratios.py` evaluates all five weights at 10% and 20% supervision.

The scripts save results after each completed run and skip configurations already recorded in their respective CSV files. To repeat training, existing result files should first be moved to a backup location.

3.5 Experimental Protocol

Experiments are conducted on Thyroid, Seismic Bumps, and a constructed synthetic dataset. Each dataset is divided into training and test subsets using a stratified 70:30 split. Min–max scaling is fitted on the training data and then applied to the test data to prevent preprocessing leakage.

Three labelled-anomaly ratios were evaluated: 5%, 10%, and 20% of the training anomalies. For each ratio, the number of labelled examples was rounded down to the nearest integer. The remaining training anomalies were treated as unlabelled observations.

The resulting labelled-anomaly counts were 3, 6, and 13 for Thyroid; 5, 11, and 23 for Seismic Bumps; and 17, 35, and 70 for the synthetic dataset, respectively.

Each combination of dataset, labelled-anomaly ratio, and bandwidth weight was evaluated using five seeds (1–5). With three datasets, three ratios, and five bandwidth weights, the complete evaluation comprised 225 runs.

Within each seed, all bandwidth configurations use the same prepared data for a given supervision level. For the additional supervision experiments, the script checks that the test data remain identical across ratios and that smaller labelled-anomaly pools are subsets of larger pools. These checks support comparisons in which the amount of supervision changes while the test set is held constant.

Performance is measured using ROC-AUC and average precision (AP). These are calculated with the `roc_auc_score` and `average_precision_score` functions from scikit-learn. The original code uses the name `aucpr` for average precision.

Results are reported as the mean and sample standard deviation across five runs. The fixed-bandwidth configuration is used as the reference baseline for evaluating the adaptive and hybrid methods.

3.6 Auxiliary LLM-as-Judge Evaluation

An auxiliary evaluation used GPT-6 in Instant mode through ChatGPT. The purpose was to examine whether an LLM could identify unusual synthetic observations from numerical evidence. It did not replace ground-truth evaluation or directly judge ADSD predictions.

Forty observations were selected from the synthetic test partition associated with split seed 1: twenty normal observations and five from each of the four anomaly groups. Selection and ordering used a separate random seed of 8240. Ground-truth labels and anomaly types were hidden from the

judge.

The prompt provided ten feature values for each candidate, together with reference quantiles and a correlation matrix derived from the training partition. The reference pool excluded the 17 revealed training anomalies in the 5% supervision setting, leaving 6,983 observations. It could still contain unlabelled anomalies.

Features were standardised by subtracting the reference median and dividing by the reference interquartile range divided by 1.349. Reference correlations were calculated after clipping standardised values to $[-3, 3]$.

The judge assigned an anomaly score from 0 to 100 and a short numerical explanation to each candidate. Scores of at least 50 were classified as anomalous. This threshold was specified before the responses were evaluated. Two responses were collected using the same prompt and model mode.

Predictions were compared with the hidden labels using accuracy, precision, recall, F1, ROC-AUC, and average precision. Repeat consistency was measured using binary decision agreement and Cohen's κ.

4 Datasets

Three datasets were used to evaluate the original and modified ADSD methods. Thyroid was selected to reproduce a result reported in the original paper. Seismic Bumps was added as a new real-world dataset, while a synthetic dataset was constructed to examine performance with different types of anomalies.

4.1 Original Benchmark Collection

The original study evaluated 25 real-world datasets covering medical, industrial, image, speech, and other anomaly detection tasks. Supplementary Table A1 reports sample counts from 1,456 to 567,498, feature counts from 3 to 400, and anomaly proportions from 0.39% to 39.91% (Jiang et al., 2023). This variation provides a relevant setting for investigating bandwidth selection.

The datasets are ALOI, annthyroid, Cardiotocography, fault, http, landsat, letter, magic.gamma, mammography, mnist, musk, optdigits, PageBlocks, pendigits, satellite, satimage-2, shuttle, skin, SpamBase, speech, thyroid, vowels, Waveform, Wilt, and yeast. This project uses Thyroid for the direct replication. Thyroid and annthyroid are separate datasets.

4.2 Thyroid

Thyroid is a medical anomaly detection dataset included in the original ADSD evaluation (Jiang et al., 2023). It contains 3,772 observations and six numerical features. Of these observations, 93 are labelled as anomalies, representing approximately 2.47% of the dataset.

Thyroid was used to check whether the original fixed-bandwidth MLP implementation could reproduce the published results. It also provides a benchmark for comparing the adaptive bandwidth configurations under the same experimental settings.

The prepared Thyroid data were obtained from the original authors' repository.[2] The dataset is stored as a NumPy `.npz` archive containing a feature matrix `X` and a binary label vector `y`. The labels distinguish normal observations ($y = 0$) from anomalies ($y = 1$).

4.3 Seismic Bumps

Seismic Bumps is a real-world dataset related to seismic activity in underground coal mines. It contains observations describing seismic and seismoacoustic measurements, together with information about mining conditions.

The prepared dataset contains 2,584 observations, including 170 anomalies (6.58%). Categorical attributes, such as seismic hazard assessments and shift information, were converted into numerical features using one-hot encoding. This produced 24 features, with no missing values in the prepared dataset.

Seismic Bumps was included to test the method on a different application domain with a higher anomaly proportion than Thyroid.

The source is the UCI Machine Learning Repository (Sikora and Wrobel, 2013). Each record summarises seismic activity during an eight-hour shift. The positive class indicates that a seismic bump with energy greater than 10^4 J occurred during the following shift. This hazardous-outcome class is treated as anomalous in this project.

The source contains 18 input attributes and one target attribute. Four categorical inputs (seismic, seismoacoustic, shift, and ghazard) were one-hot encoded into ten indicator columns. Together with 14 numerical inputs, this produced the 24 prepared features. The target was excluded from the input

[2] https://github.com/Minqi824/Overlap/tree/main/datasets

matrix. The source data contain no missing values.

4.4 Constructed Synthetic Dataset

A synthetic dataset was created to examine the behaviour of bandwidth selection under controlled conditions. It contains 10,000 observations with 10 numerical features. A total of 500 observations (5%) are anomalies.

The anomalies were divided equally into four categories, with 125 observations per category:

- **Global anomalies:** Observations with feature values that differ substantially from the main data distribution.
- **Local anomalies:** Variance-inflated observations generated around the normal centre. This group is called local in the generator, although local-density abnormality is not guaranteed by construction.
- **Dependency anomalies:** Observations that violate relationships between features.
- **Cluster anomalies:** Groups of observations that form unusual patterns away from the main normal population.

The synthetic data were generated using a fixed random seed of 42. Including several anomaly types allows the experiment to examine a broader range of abnormal patterns than a dataset containing only one anomaly type.

Generation parameters. The 9,500 normal observations were sampled from a zero-mean, ten-dimensional Gaussian distribution. A positive-definite matrix generated with seed 42 was normalised to unit diagonal to define its covariance matrix Σ.

For feature j, let m_j be the maximum normal value and r_j the normal-data range. Global anomalies were sampled uniformly from $[m_j + 0.5r_j,\ m_j + 1.5r_j]$ in each feature. The local group used the same zero mean and covariance 3Σ.

Dependency anomalies were constructed by selecting 125 normal records and independently permuting their feature columns. This preserves their empirical marginal values while disrupting dependencies between features. Cluster anomalies were sampled with mean $(3, \ldots, 3)$ and covariance 0.25Σ.

Each anomaly group contained 125 observations. All observations were shuffled after generation, while keeping features, binary labels, and anomaly-type metadata aligned. Labels were assigned automatically from the generation process; no human annotation was performed.

Dataset	x_1	x_2	x_3	y
Thyroid	0.7742	0.0011	0.1376	0
Thyroid	0.2473	0.0005	0.2799	0
Seismic	15180	48	-72	0
Seismic	14720	33	-70	0
Synthetic	-0.8076	0.4244	-0.3735	0
Synthetic	1.1938	0.2512	0.4529	0

Table 1: Examples from the prepared dataset archives. Values are rounded where necessary. These first records illustrate the data format; they do not represent the anomaly classes.

Examples of prepared data. Table 1 shows the first two stored records from each dataset, before the experimental training pipeline applies min–max scaling. Only the first three feature columns are displayed. Positional indices are used consistently across datasets. The target label is stored separately from the input features.

4.5 Dataset Summary

Table 2 summarises the datasets used in the experiments.

Dataset	Samples	Features	Anomalies
Thyroid	3,772	6	93
Seismic Bumps	2,584	24	170
Synthetic	10,000	10	500

Table 2: Summary of the datasets used in the experiments. Feature counts refer to the prepared datasets.

All three datasets followed the same stratified 70:30 training and test split described in Section 3. The training data were scaled using min–max normalisation. Separate experiments retained 5%, 10%, or 20% of training anomalies as labelled examples, with the remaining training anomalies treated as unlabelled.

5 Results

This section compares the original ADSD results with the replication and evaluates the effect of KDE bandwidth selection. Each configuration was tested using five random seeds. Results are presented as mean $\pm$ sample standard deviation.

5.1 Replication of the Original ADSD Results

The first experiment reproduced the fixed-bandwidth MLP configuration on the Thyroid dataset. The original study reported a ROC-AUC of 0.981 and an AUC-PR of 0.814 for MLP-Overlap with 5% labelled anomalies (Jiang et al., 2023).

The replication achieved a mean ROC-AUC of 0.9811 ± 0.0170 and average precision (AP) of 0.8138 ± 0.0394 across five seeds. Table 3 compares the results.

Source	ROC-AUC	AUC-PR / AP
Original ADSD	0.981	0.814
Replication	0.9811	0.8138

Table 3: Published ADSD results and fixed-bandwidth replication on Thyroid. Replication values are five-seed means.

The difference in ROC-AUC was only $+0.0001$. The numerical difference between the published AUC-PR and the replicated AP was -0.0002. The inspected original implementation also uses `average_precision_score`, although the metric is named `aucpr`. Thus, the code-based comparison uses average precision in both cases. The results are very close.

5.2 Bandwidth Ablation Results

Five interpolation weights were evaluated on Thyroid, Seismic Bumps, and the synthetic dataset with 5% labelled training anomalies. Table 4 presents the results for both evaluation metrics.

Dataset	α	ROC-AUC	AP
Thyroid	0.0	$\mathbf{0.9811 \pm 0.0170}$	$\mathbf{0.8138 \pm 0.0394}$
	0.2	0.9719 ± 0.0312	0.8091 ± 0.0509
	0.5	0.9627 ± 0.0408	0.7872 ± 0.0696
	0.8	0.9080 ± 0.1156	0.6833 ± 0.2244
	1.0	0.8495 ± 0.1645	0.4784 ± 0.3483
Seismic	0.0	$\mathbf{0.5272 \pm 0.0728}$	$\mathbf{0.0951 \pm 0.0413}$
	0.2	0.5153 ± 0.0543	0.0914 ± 0.0391
	0.5	0.4939 ± 0.0672	0.0861 ± 0.0320
	0.8	0.4889 ± 0.0841	0.0903 ± 0.0464
	1.0	0.4867 ± 0.1018	0.0874 ± 0.0392
Synthetic	0.0	$\mathbf{0.7362 \pm 0.0483}$	0.5219 ± 0.1273
	0.2	0.7157 ± 0.0625	0.5217 ± 0.1496
	0.5	0.7249 ± 0.0472	0.5158 ± 0.1013
	0.8	0.7350 ± 0.0346	$\mathbf{0.5508 \pm 0.0885}$
	1.0	0.7152 ± 0.0516	0.5337 ± 0.1162

Table 4: Bandwidth ablation results with 5% labelled training anomalies across five random seeds (mean $\pm$ sample standard deviation).

On Thyroid, the fixed bandwidth ($\alpha = 0$) achieved the highest mean ROC-AUC and AP. Performance decreased as the adaptive component increased. The fully adaptive configuration ($\alpha = 1$) also showed greater variation between seeds.

Seismic Bumps produced lower scores across all configurations. The fixed bandwidth again gave the highest mean ROC-AUC and AP, although differences in AP were relatively small.

For the synthetic dataset, the highest mean AP was obtained at $\alpha = 0.8$, increasing from 0.5219 to 0.5508. However, the fixed bandwidth retained the highest mean ROC-AUC at 0.7362.

Figures 1 and 2 show the changes in performance across interpolation weights.

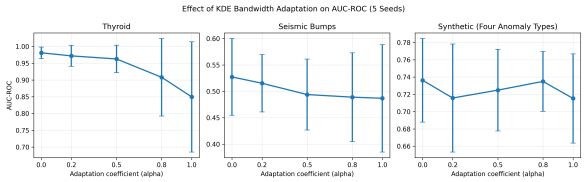


Figure 1: Mean ROC-AUC across five seeds for different KDE bandwidth interpolation weights with 5% labelled training anomalies. Error bars show standard deviations.

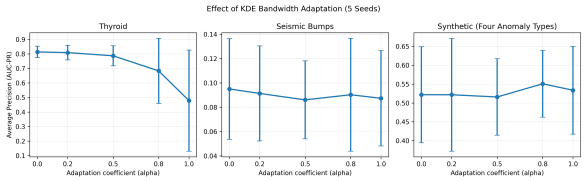


Figure 2: Mean average precision across five seeds for different KDE bandwidth interpolation weights with 5% labelled training anomalies. Error bars show standard deviations.

5.3 Effect of Labelled-Anomaly Availability

Additional experiments evaluated 10% and 20% labelled training anomalies using the same five bandwidth weights and five random seeds. Table 5 compares the fixed bandwidth with the $\alpha = 0.8$ hybrid setting across all three supervision levels. Complete results for all weights are provided in the appendix.

On Seismic Bumps, the fixed-bandwidth ROC-AUC increased from 0.5272 at 5% supervision to 0.6015 at 10% and 0.6594 at 20%. Its AP also increased from 0.0951 to 0.1253 and 0.1535. However, the hybrid method did not consistently outperform the fixed baseline.

On Thyroid, additional supervision reduced the large performance deterioration observed with fully adaptive bandwidth at 5%. The highest mean AP

was obtained with $\alpha = 0.8$ at 10% and $\alpha = 1.0$ at 20%. Performance did not increase monotonically with supervision: fixed-bandwidth AP was 0.8701 at 10% and 0.8594 at 20%.

On the synthetic dataset, $\alpha = 0.8$ achieved the highest mean AP at all three supervision levels. At 10% and 20%, it also achieved the highest mean ROC-AUC among the tested weights. These findings indicate that bandwidth effects depend on both the dataset and the availability of labelled anomalies.

5.4 Statistical Comparison

An exploratory paired comparison was conducted between the fixed bandwidth ($\alpha = 0$) and the hybrid configuration ($\alpha = 0.8$) on the synthetic dataset with 5% labelled training anomalies. The same five seeds were used for both configurations.

The mean AP difference was $+0.0289$. A paired t-test gave a p-value of approximately 0.267, with a 95% confidence interval of $[-0.0334, 0.0913]$.

The confidence interval includes zero, and the p-value is above 0.05. Therefore, the observed improvement is not statistically significant based on these five runs.

The exploratory synthetic-data comparison was also repeated at the additional supervision levels. At 10%, the mean AP difference between $\alpha = 0.8$ and the fixed baseline was $+0.0498$, with a two-sided paired-test p-value of 0.340 and a 95% confidence interval of $[-0.0780, 0.1775]$. At 20%, the difference was $+0.0223$, with $p = 0.082$ and a confidence interval of $[-0.0045, 0.0492]$.

None of these comparisons established an AP improvement at the 5% significance level. These are exploratory comparisons without a multiple-comparison adjustment.

Overall, adaptive bandwidth selection did not consistently improve ADSD performance across datasets and supervision levels. Although some adaptive configurations achieved higher mean scores, the exploratory synthetic-data comparisons did not establish an AP improvement at the 5% significance level.

5.5 Auxiliary LLM-as-Judge Results

Both responses contained valid scores and decisions for all 40 observations. Table 6 summarises performance. The two responses agreed on 35 of 40 binary decisions, giving 87.5% agreement and Cohen's $\kappa = 0.7475$.

Both responses identified all five global and all five cluster anomalies. Each detected four of the five variance-inflated local anomalies. Dependency anomalies were more difficult: the first response detected one of five, while the second detected none.

These observations suggest that the judge was more successful on large numerical departures than on dependency violations in this small sample. With only five examples per anomaly type, these results are descriptive and do not establish general performance across anomaly categories.

6 Discussion and Limitations

6.1 Effect of Adaptive Bandwidth Selection

The effect of bandwidth selection depended on both the dataset and the labelled-anomaly ratio. At 5% supervision, the fixed bandwidth achieved the highest mean ROC-AUC and AP on Thyroid. The fully adaptive configuration showed a large performance decrease and greater variation between seeds. With additional labelled anomalies, this deterioration was substantially reduced. The highest mean AP was obtained with $\alpha = 0.8$ at 10% and $\alpha = 1.0$ at 20%. These descriptive differences do not establish a reliable advantage over the fixed baseline.

On Seismic Bumps, increasing supervision improved the fixed-bandwidth results: ROC-AUC rose from 0.5272 at 5% to 0.6594 at 20%, while AP rose from 0.0951 to 0.1535. Bandwidth adaptation did not provide a consistent additional benefit.

On the synthetic dataset, $\alpha = 0.8$ achieved the highest mean AP at all three supervision levels. It also achieved the highest mean ROC-AUC at 10% and 20%. However, its AP differences from the fixed baseline were not statistically significant at the 5% level.

One possible explanation for unstable adaptive performance is sensitivity to narrow bandwidths and repeated labelled examples within mini-batches. Bandwidth values were not recorded during the reported experiments, so this explanation remains unconfirmed. The findings apply to the tested clipped bandwidth rule rather than adaptive KDE methods in general.

6.2 Limitations

Several limitations should be considered when interpreting the results.

The proposal also planned at least two additional real-world datasets; the completed evaluation in-

Dataset	Labels	Fixed ROC-AUC	Hybrid ROC-AUC	Fixed AP	Hybrid AP
Thyroid	5%	0.9811 ± 0.0170	0.9080 ± 0.1156	0.8138 ± 0.0394	0.6833 ± 0.2244
Thyroid	10%	0.9941 ± 0.0026	0.9912 ± 0.0096	0.8701 ± 0.0261	0.8760 ± 0.0131
Thyroid	20%	0.9937 ± 0.0059	0.9867 ± 0.0212	0.8594 ± 0.1001	0.8386 ± 0.1015
Seismic	5%	0.5272 ± 0.0728	0.4889 ± 0.0841	0.0951 ± 0.0413	0.0903 ± 0.0464
Seismic	10%	0.6015 ± 0.0507	0.5685 ± 0.0690	0.1253 ± 0.0393	0.1113 ± 0.0319
Seismic	20%	0.6594 ± 0.0683	0.6539 ± 0.0731	0.1535 ± 0.0412	0.1478 ± 0.0377
Synthetic	5%	0.7362 ± 0.0483	0.7350 ± 0.0346	0.5219 ± 0.1273	0.5508 ± 0.0885
Synthetic	10%	0.7583 ± 0.0604	0.7709 ± 0.0394	0.5656 ± 0.0874	0.6153 ± 0.0500
Synthetic	20%	0.7984 ± 0.0516	0.8084 ± 0.0427	0.6198 ± 0.0361	0.6422 ± 0.0389

Table 5: Fixed bandwidth ($\alpha = 0$) and hybrid bandwidth ($\alpha = 0.8$) across supervision levels. Values are mean $\pm$ sample standard deviation over five seeds. The hybrid weight is held constant across these comparisons; it is not the best-performing weight in every setting.

Metric	Response 1	Response 2
Accuracy	0.7750	0.8000
Precision	0.7895	0.8750
Recall	0.7500	0.7000
F1	0.7692	0.7778
ROC-AUC	0.8363	0.8413
AP	0.8829	0.8913

Table 6: Auxiliary GPT-6 Instant judgments on the same 40 synthetic test observations. Binary predictions use a score threshold of 50.

cludes only Seismic Bumps alongside constructed synthetic data.

First, the experiments used only the MLP architecture. Although three labelled-anomaly ratios were evaluated, the results cannot be generalised to the other architectures examined in the original ADSD study (Jiang et al., 2023).

Second, only three datasets and five random seeds were used. This limits the statistical power of the comparisons, particularly for the small improvement observed on the synthetic dataset.

Third, the interpolation weights were compared using test-set results without a separate validation-based selection procedure. Consequently, the best-performing weight should be treated as an experimental observation rather than a validated parameter choice.

Fourth, the adaptive bandwidth is calculated from the current mini-batch. The method may therefore be sensitive to batch composition and score variation. Bandwidth values were not logged during training, which prevents a detailed analysis of their stability.

Finally, the synthetic dataset contains four constructed anomaly types, but performance was evaluated using overall ROC-AUC and AP. The results do not establish which anomaly type benefited most from adaptive bandwidth selection. The constructed patterns may also differ from anomalies found in real-world data.

The auxiliary LLM evaluation used a small, deliberately balanced sample containing 50% anomalies, rather than the synthetic dataset's 5% anomaly prevalence. Its AP and precision therefore cannot be directly compared with ADSD results on the full test partition. The two responses measure repeat consistency, not agreement between independent human annotators. The judge had access only to numerical summaries and provided no independent expert validation of the data or the bandwidth extension.

6.3 Future Work

Future experiments could evaluate additional datasets, neural network architectures, and labelled-anomaly ratios. Recording the bandwidth values during training would help explain how adaptive KDE behaves across different datasets and mini-batches.

A separate validation set could also be used to select the interpolation weight before final test evaluation. In addition, measuring performance separately for each synthetic anomaly type could provide a clearer understanding of when adaptive bandwidth selection is useful.

7 Ethical Considerations

Anomaly detection can support decisions in sensitive areas such as healthcare and industrial safety. However, incorrect predictions may have serious consequences. False positives can lead to unnecessary investigations, while false negatives may allow important anomalies to remain undetected.

The ethical assessment also considers research

integrity. The Australian Code for the Responsible Conduct of Research identifies honesty, rigour, transparency, and fairness among its guiding principles (National Health and Medical Research Council et al., 2018). For this project, these principles require accurate reporting of both positive and negative results, clear documentation of experimental choices, and appropriate acknowledgement of the original work and data sources.

7.1 Risks of Incorrect Predictions

The Thyroid dataset is related to medical anomaly detection, where incorrect predictions could affect the identification of unusual patient records. Similarly, Seismic Bumps is associated with mining conditions, where missed anomalies could have safety implications.

The relatively low ROC-AUC results on Seismic Bumps show that the tested model is not reliable enough for safety-critical decisions. The experimental results should therefore not be treated as evidence that ADSD is suitable for real-world deployment in these settings.

7.2 Data Privacy and Representation

Medical and industrial datasets may contain sensitive information. Although this project uses prepared research datasets, data privacy remains important when applying similar methods to real-world records. Access controls, data minimisation, and appropriate handling of personal information would be necessary in a practical application.

The synthetic dataset avoids using real personal records and allows different anomaly patterns to be studied under controlled conditions. However, its generated observations may not represent the complexity of real-world anomalies. Conclusions drawn from synthetic data should therefore be interpreted carefully.

7.3 Responsible Reporting and Reproducibility

All bandwidth configurations were evaluated using the same experimental settings and five random seeds. Both positive and negative results are reported to avoid presenting the adaptive method as consistently better than the baseline.

The synthetic-data comparisons did not establish an AP improvement at the 5% significance level. The comparisons are exploratory, and interpolation weights were not selected using an independent validation set. These limitations are reported to avoid overstating the effectiveness of the extension.

For responsible use, anomaly detection systems should be evaluated on representative data and monitored after deployment. In high-risk applications, model predictions should support human review rather than replace professional judgement.

7.4 Open Research and Attribution

The original ADSD work provides source code and benchmark data, enabling direct investigation of its reported results. This availability supported the replication and allowed the bandwidth extension to be compared with the original implementation.

The original source code is distributed under the MIT licence. Its copyright and licence notice must be retained when redistributing the implementation. The project distinguishes the original authors' contributions from the bandwidth modification and additional experiments.

Seismic Bumps is distributed under the Creative Commons Attribution 4.0 licence (Sikora and Wrobel, 2013). Its use requires appropriate attribution. Public availability does not imply that all datasets share the same licence or that redistribution is unrestricted.

The synthetic dataset contains generated observations rather than real personal records. Providing its generation script, parameters, and labels supports inspection and reuse. Sharing individual run results also allows reported averages and variability to be checked independently, supporting transparent research (National Health and Medical Research Council et al., 2018).

8 Conclusion

The selected ADSD MLP configuration closely reproduced the published Thyroid results, achieving ROC-AUC of 0.9811 and AP of 0.8138 at 5% supervision.

Across 225 runs, the tested bandwidth extension showed no consistent advantage over the fixed baseline. On synthetic data, $\alpha = 0.8$ achieved the highest mean AP at all three supervision levels, but paired comparisons did not establish statistically significant improvements.

The auxiliary LLM evaluation suggested difficulty identifying dependency anomalies, but did not validate the bandwidth extension. Further work should examine validation-based bandwidth selection and additional datasets and architectures.

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A Complete Bandwidth Results

Labels	α	ROC-AUC	AP
5%	0.0	0.9811 ± 0.0170	0.8138 ± 0.0394
5%	0.2	0.9719 ± 0.0312	0.8091 ± 0.0509
5%	0.5	0.9627 ± 0.0408	0.7872 ± 0.0696
5%	0.8	0.9080 ± 0.1156	0.6833 ± 0.2244
5%	1.0	0.8495 ± 0.1645	0.4784 ± 0.3483
10%	0.0	0.9941 ± 0.0026	0.8701 ± 0.0261
10%	0.2	0.9941 ± 0.0024	0.8657 ± 0.0219
10%	0.5	0.9942 ± 0.0023	0.8680 ± 0.0205
10%	0.8	0.9912 ± 0.0096	0.8760 ± 0.0131
10%	1.0	0.9934 ± 0.0022	0.8667 ± 0.0316
20%	0.0	0.9937 ± 0.0059	0.8594 ± 0.1001
20%	0.2	0.9934 ± 0.0071	0.8627 ± 0.0998
20%	0.5	0.9892 ± 0.0163	0.8523 ± 0.1099
20%	0.8	0.9867 ± 0.0212	0.8386 ± 0.1015
20%	1.0	0.9925 ± 0.0108	0.8770 ± 0.0744

Table 7: Complete bandwidth results for Thyroid: mean $\pm$ sample standard deviation over five seeds.

Labels	α	ROC-AUC	AP
5%	0.0	0.5272 ± 0.0728	0.0951 ± 0.0413
5%	0.2	0.5153 ± 0.0543	0.0914 ± 0.0391
5%	0.5	0.4939 ± 0.0672	0.0861 ± 0.0320
5%	0.8	0.4889 ± 0.0841	0.0903 ± 0.0464
5%	1.0	0.4867 ± 0.1018	0.0874 ± 0.0392
10%	0.0	0.6015 ± 0.0507	0.1253 ± 0.0393
10%	0.2	0.5972 ± 0.0560	0.1274 ± 0.0414
10%	0.5	0.5884 ± 0.0515	0.1152 ± 0.0321
10%	0.8	0.5685 ± 0.0690	0.1113 ± 0.0319
10%	1.0	0.5807 ± 0.0822	0.1195 ± 0.0427
20%	0.0	0.6594 ± 0.0683	0.1535 ± 0.0412
20%	0.2	0.6597 ± 0.0695	0.1526 ± 0.0432
20%	0.5	0.6427 ± 0.0767	0.1429 ± 0.0373
20%	0.8	0.6539 ± 0.0731	0.1478 ± 0.0377
20%	1.0	0.6587 ± 0.0908	0.1468 ± 0.0390

Table 8: Complete bandwidth results for Seismic Bumps: mean $\pm$ sample standard deviation over five seeds.

Labels	α	ROC-AUC	AP
5%	0.0	0.7362 ± 0.0483	0.5219 ± 0.1273
5%	0.2	0.7157 ± 0.0625	0.5217 ± 0.1496
5%	0.5	0.7249 ± 0.0472	0.5158 ± 0.1013
5%	0.8	0.7350 ± 0.0346	0.5508 ± 0.0885
5%	1.0	0.7152 ± 0.0516	0.5337 ± 0.1162
10%	0.0	0.7583 ± 0.0604	0.5656 ± 0.0874
10%	0.2	0.7706 ± 0.0428	0.5986 ± 0.0760
10%	0.5	0.7601 ± 0.0312	0.5672 ± 0.0542
10%	0.8	0.7709 ± 0.0394	0.6153 ± 0.0500
10%	1.0	0.7508 ± 0.0325	0.5880 ± 0.0505
20%	0.0	0.7984 ± 0.0516	0.6198 ± 0.0361
20%	0.2	0.8005 ± 0.0543	0.6319 ± 0.0370
20%	0.5	0.7951 ± 0.0473	0.5978 ± 0.0559
20%	0.8	0.8084 ± 0.0427	0.6422 ± 0.0389
20%	1.0	0.7943 ± 0.0462	0.6282 ± 0.0337

Table 9: Complete bandwidth results for Synthetic: mean $\pm$ sample standard deviation over five seeds.